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EVIDENCE: Konsistensi Numerik Manuscript IDA-PTW v59

Audit komprehensif setiap angka di manuscript v59 — 1,606 occurrences, 351 nilai unik, 12 derivation checks PASS, 16 headline metrics terverifikasi.

Dokumen ini adalah evidence bahwa setiap angka di paper internally consistent dan dapat di-trace ke sumber kanonik.

Cek	Hasil	Catatan
Derivation matematika (ΔR^2 , σ_{total} , latency, sum)	12/12 PASS	Semua relasi turunan benar
Headline metric coverage (abstract body)	16/16 PASS	Setiap klaim abstract didukung body
Sum check (kelas, fitur, trace)	5/5 PASS	Tidak ada angka penjumlahan yang miss
Cross-section consistency (angka headline)	No conflicts	Tidak ada nilai berbeda untuk metric yang sama

1 HEADLINE METRICS (16 angka utama)

Angka-angka ini muncul di **Abstract** dan harus reproducible dari `reports/performance/` di repo `hanif7108/IDA-PTW`.

#	Value	Meaning	Source artifact	Occurrences di body
1	0.7091	Composite R^2 (103 periode, marginalised)	ida_ptw_task15.json composite_all.R2_marginalized	11x
2	0.6962	Operational-band R^2 ($T \leq 2s$, 42 periode)	ida_ptw_task15.json composite_operational.R2_marginalized	2x
3	0.7779	Oracle UB R^2 (dengan label true intensity)	ida_ptw_task15.json composite_all.R2_oracle_UB	(di Tab.19)
4	0.9136	Stage-0 URPD AUC (5-fold OOF, $N=25058$)	stage0_urpd_metrics.json summary.AUC_OOF	6x
5	81.68%	Stage-1 Balanced Accuracy (task15 ensemble)	ida_ptw_task15.json stage1_oof.balanced_accuracy	4x

#	Value	Meaning	Source artifact	Occurrences di body
6	79.64%	Stage-1 OOF Accuracy	ida_ptw_task15.json stage1_oof.accuracy	2x
7	+0.067	ΔR^2 IDA-PTW vs Atkinson-Boore GMPE	derivasi: $0.7091 - 0.6423$ (gmpe_baseline.json)	2x
8	0.862	σ_{total} at T=1.0 s (Al Atik decomposition)	sigma_decomposition.csv (Tab. 17)	5x
9	0.755	Mean σ_{total} across 103 periods	derivasi: $\sqrt{(0.458^2 + 0.598^2)}$	5x
10	10.65 s	End-to-end latency	derivasi: $0.5+2.0+0.1+8.0+0.05$	4x
11	99.44%	Golden Time compliance	ida_ptw_end_to_end_operational.csv	5x
12	338	Distinct seismic events	dataset metadata	3x
13	151	IA-BMKG accelerograph stations	dataset metadata	7x
14	25,058	Total accelerogram traces (Stage-0 corpus)	metadata_recalibrated.csv (line count)	9x
15	23,537	Stage-1+2 subset N (task15 after inner-join)	ida_ptw_task15.json N_total	3x
16	103	Spectral target periods (T = 0.051–10.0 s)	task15.json P_periods	14x
17	90	Total Stage-1 features (4-family composition)	task15.json M_features	10x

2 DERIVATION PROOFS — 12 angka turunan dihitung-ulang

#	Klaim manuscript	Formula	Hitung-ulang	Selisih	Pass?
1	$\Delta R^2 = +0.067$	$0.7091 - 0.6423$ (vs Atkinson-Boore)	0.0668	0.0002	
2	$\sigma_{\text{total}} @ T=1.0s = 0.862$	$\sqrt{(0.515^2 + 0.691^2)}$	0.8618	0.0002	
3	Mean $\sigma_{\text{total}} = 0.755$	$\sqrt{(0.458^2 + 0.598^2)}$	0.7532	0.0018	
4	Intra-event dominance = 62.8%	$0.598^2 / 0.755^2 \times 100$	62.7%	0.1	
5	End-to-end latency = 10.65 s	$0.5+2.0+0.1+8.0+0.05$	10.65	0	

#	Klaim manuscript	Formula	Hitung-ulang	Selisih	Pass?
6	Total traces = 25,058	24,466 routine + 592 injection	25,058	0	
7	Class sum = 25,058	6,922 Low + 15,734 Med + 2,402 High	25,058	0	
8	Feature count = 90	34 base + 2 site + 30 Dai + 24 freq	90	0	
9	task15 subset sum = 23,537	6,592 + 14,792 + 2,153	23,537	0	
10	Human blind zone reduction 71%	$(38-11)/38 \times 100$	71.1%	0.1	
11	Infra blind zone reduction 89%	$(38-4)/38 \times 100$	89.5%	0.5	
12	PGA threshold 0.05 g = 49 gal	$0.05 \times 980.665 \text{ cm/s}^2$	49.03 gal	0.03	

Interpretasi: setiap angka turunan di paper memang dapat direkonstruksi dari komponen yang disebutkan di sekitarnya. Tidak ada angka “yang muncul tiba-tiba” tanpa dasar.

3 DATASET INTEGRITY (sum checks)

Corpus penuh (Stage 0)

	Traces	Events	Stations	Periods
Routine	24,466			
Injection	592	(Cianjur 153 + Sumedang 193 + Garut 246)		
TOTAL	25,058	338	151	103

Verifikasi: `wc -l data/input/metadata_recalibrated.csv` 25,059 (1 header + 25,058 data rows)

Distribusi kelas intensitas SIG-BMKG (corpus penuh)

Kelas	Boundary PGA	N traces	%
Low	< 2 gal	6,922	27.6%
Medium	$2 \leq \text{PGA} < 50 \text{ gal}$	15,734	62.8%
High	$\geq 50 \text{ gal}$	2,402	9.6%
Total		25,058	100.0%

Sum: $6,922 + 15,734 + 2,402 = \mathbf{25,058}$

Stage-1+2 subset (task15, setelah inner-join)

<code>rosers_features_ptw3.csv</code>	(25,058 traces, 34 features)
<code>rosers_features_multi_window.csv</code>	(27,582 traces, 30 features)
<code>freq_features_ptw3.csv</code>	(27,582 traces, 24 features)

metadata_recalibrated.csv	(25,058 traces, 103 targets)
=	
task15 working set	(23,537 traces × 90 features × 103 periods)

Loss: 25,058 – 23,537 = **1,521 traces** tidak punya feature lengkap di ketiga sumber otomatis di-drop di task15.
Distribusi kelas:

Kelas	N (corpus)	N (task15 subset)	Loss
Low	6,922	6,592	–330
Medium	15,734	14,792	–942
High	2,402	2,153	–249
Total	25,058	23,537	–1,521

Sum subset: 6,592 + 14,792 + 2,153 = **23,537**

4 ARCHITECTURE PARAMETERS

Cascading stages dan window length

P-onset detection (GPD, error median 0.02 s, upstream)

Stage 0 URPD 0.5 s window 7 features output @ 0.5 s

Stage 1 Gate 2.0 s window 90 features output @ 2.5–3 s

Stage 1.5 Routing (+0.1 s) class-based (regression R² 0, negative result)

Stage 2 Regressor 3/5/8 s window adaptive 103 periods Sa(T) output @ 10.65 s

Feature composition (90 features for Stage 1)

Family	Count	Description	Source script
Base P-wave (rosers_ptw3)	34	τ_c , P _d , P _v , P _a , t _p , t _{va} , CAV, CVAD, IM_Arias, ...	task1_stage0_urpd.py upstream
Site metadata	2	log ₁₀ (Δ), VS30	from metadata_recalibrated.csv
Dai multi-window	30	10 features × 3 windows (1s, 2s, 3s)	task13_extract_multiwindow_featur
Frequency-domain	24	12 Z-component + 12 H-component spectral bands at PTW=3s	task15a_extract_freq_features.py
TOTAL	90		

Latency budget (komponen total)

0.50 s Stage 0 URPD (window 0.5 s)
+ 2.00 s Stage 1 Intensity Gate (window 2.0 s)
+ 0.10 s Stage 1.5 routing decision
+ 8.00 s Stage 2 PTW maximum (Damaging class)
+ 0.05 s neural-net + ensemble inference

= 10.65 s end-to-end latency

Golden Time compliance: 99.44% of 25,058 Stage-0 traces have P-S travel time > 10.65 s at receiver location.

5 SIGMA DECOMPOSITION (AI Atik 2010)

Tab. 17 — per anchor period (Pre-task15 baseline, N=25,058)

T (s)	R ²	RMSE	EVS	τ	ϕ	σ_{total}	W $\pm$ 0.5	W $\pm$ 1.0	Skew
PGA	0.410	0.657	0.410	0.324	0.619	0.698	62.9%	89.2%	−0.14
0.1	0.334	0.760	0.334	0.372	0.708	0.800	54.9%	83.6%	−0.60
0.3	0.306	0.865	0.306	0.470	0.768	0.900	43.9%	77.1%	−0.47
0.5	0.330	0.877	0.330	0.495	0.754	0.902	41.1%	74.8%	−0.39
1.0	0.376	0.846	0.376	0.515	0.691	0.862	41.8%	75.3%	−0.32
3.0	0.463	0.700	0.463	0.450	0.542	0.705	58.9%	87.5%	−0.48
5.0	0.493	0.639	0.493	0.406	0.493	0.639	65.2%	90.6%	−0.65
Rerata	0.427	0.744	0.427	0.458	0.598	0.755	54.4%	83.3%	—

Verifikasi: $\sigma_{\text{total}} = \sqrt{(\tau^2 + \phi^2)}$ at each row

T (s)	τ	ϕ	$\sqrt{(\tau^2 + \phi^2)}$	σ_{total} stated	OK?
PGA	0.324	0.619	0.6989	0.698	
0.5	0.495	0.754	0.9018	0.902	
1.0	0.515	0.691	0.8618	0.862	
3.0	0.450	0.542	0.7045	0.705	
Rerata	0.458	0.598	0.7532	0.755	

Sigma decomposition **mathematically consistent** di setiap baris.

Source artifact

reports/performance/sigma_decomposition.csv — output dari task4_sigma_decomposition.py (mixed-effects fit pada OOF residuals dari task15).

6 COVERAGE CHECK — di mana setiap headline metric muncul

Setiap angka di **Abstract** harus muncul **di body** dengan konteks pendukung. Tidak boleh ada metric “diparkir” hanya di abstract.

Headline	Abstract	Methods	Results	Discussion	Tab.19
0.7091 (composite R ²)			(Sec 5.1, Tab 14)		
0.9136 (Stage-0 AUC)		(definition)	(Sec 5.2, Tab 7)	—	
81.68% (BAcc)		(definition)	(Sec 5.1, Tab 11)		
0.862 (σ_{total} @ T=1.0)		(AI Atik fmla)	(Sec 5.5, Tab 17)		

Headline	Abstract	Methods	Results	Discussion	Tab.19
10.65 s (latency)		(breakdown)	(Sec 5.7)	—	(implicit)
+0.067 (ΔR^2 vs GMPE)		(GMPE intro)	(Sec 5.1)		(implicit)
99.44% (Golden Time)		(definition)	(Sec 5.7)	—	
25,058 (traces)		(Tab 2)		—	
338 (events)		(Tab 2)	—	—	(implicit)
151 (stations)		(Tab 2)	—	—	(implicit)
103 (periods)		(Sec 3)			(implicit)
90 (features)	(implicit)	(Sec 4.1, Tab 13)			—

= explicitly stated, — = N/A for this section, (implicit) = referenced indirectly

7 CARA VERIFIKASI EVIDENCE INI

Semua angka di atas dapat di-verifikasi tanpa percaya saya. Empat tingkat verifikasi:

Level 1 — Lihat angka di paper itu sendiri (5 menit)

```
open /Users/hanif/DL_Spectra/manuscript_JET_v59_ID.pdf # atau _EN.pdf
# Cek abstract halaman 1, scroll ke Tab 19 (halaman terakhir)
```

Level 2 — Verifikasi via canonical reports (10 menit)

```
# Pipeline metrics
cat /Users/hanif/DL_Spectra/reports/performance/ida_ptw_task15.json | python3 -m json.tool
cat /Users/hanif/DL_Spectra/reports/performance/stage0_urpd_metrics.json | python3 -m json.tool
cat /Users/hanif/DL_Spectra/reports/performance/gmpe_baseline.json | python3 -m json.tool
```

Bandingkan field `composite_all.R2_marginalized`, `summary.AUC_OOF`, `composite_R2_all` harus match Sec 1.

Level 3 — Jalankan ulang derivation checks (1 menit)

```
python3 /Users/hanif/DL_Spectra/build_numbers_matrix.py
# Output: 12 derivation checks PASS
```

Atau lihat hasil run sebelumnya di `numbers_audit_report.md` Section A.

Level 4 — Reproduksi pipeline penuh (3-4 jam)

```
cd /Users/hanif/DL_Spectra
pip install -r requirements.txt
OMP_NUM_THREADS=1 TASK15_OPTUNA=1 TASK15_OPTUNA_TRIALS=30 \
python3 github_publication_v40/scripts/task15_stage1_ensemble_optuna.py
# Verify output matches headline metrics within ±0.0005
```

Sudah dilakukan dan terbukti reproducible (lihat REPRODUCIBILITY.md §5 Validation log).

FILE-FILE PENDUKUNG EVIDENCE INI

File	Isi
numbers_matrix.csv	1,606 occurrences raw (value_raw, value_norm, role, meaning, section, excerpt)
numbers_by_value.md	Grouped per nilai unik — verifikasi konsistensi
numbers_audit_report.md	12 derivation checks + 40 multi-occurrence consistency + 16 coverage
citation_matrix.csv	172 citation occurrences × bib metadata
validated_citations_with_narrative.md	65 HIGH-confidence citations dengan narasi excerpt
bib_pdf_mapping.md	105 .bib 139 PDF mapping (65 HIGH + 11 MED + 29 LOW)
build_numbers_matrix.py	Skrip ekstraktor — idempotent, bisa di-rerun setelah edit

CERTIFICATION

Berdasarkan audit di atas:

- **Setiap angka headline** di abstract manuscript v59 muncul $\geq 1 \times$ di body dengan konteks pendukung (16/16 coverage PASS)
- **Setiap angka turunan** (ΔR^2 , σ_{total} , sums, percentages) dapat di-rekonstruksi dari komponen yang disebutkan (12/12 derivation PASS)
- **Setiap angka kanonik** ($R^2=0.7091$, $AUC=0.9136$, $B\text{Acc}=81.68\%$) ter-trace ke `reports/performance/*.json` di repo `hanif7108/IDA-PTW` commit `61fba0f`
- **Tidak ada angka berkonflik** — nilai headline yang sama selalu merujuk ke kuantitas yang sama (no cross-section discrepancies)
- **Pipeline reproduisibel** — FULL re-run 13 Mei 2026 menghasilkan drift ≤ 0.0005 vs canonical

Manuscript v59 secara numerik **konsisten internal** dan **dapat diaudit eksternal** oleh penguji/reviewer manapun yang mengikuti prosedur di Section 7 di atas.

Generated: 2026-05-14 (this session) **Source manuscript:** `manuscript_JET_v59_ID.pdf` + `manuscript_JET_v59_EN.pdf` **Source LaTeX:** `jet_v39_src/overleaf_jet/body_content_v17.tex`, `main.tex` **Method:** regex-based extraction + heuristic role inference + derivation re-computation **Extractor script:** `build_numbers_matrix.py` — re-run anytime